HALF YEARLY EXAMINATION -2025

CLASS: 12 **PHYSICS** **Reg.No**

Time : 3.00 Hours **MARKS : 70**

PART – I

$15 \times 1 = 15$

Note: **(i) Answer all the questions**
(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer

1. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10 Ω is
a) 0.2 Ω b) 0.5 Ω c) 0.8 Ω d) 1.0 Ω

2. A surface $S = 10\,\hat{j}$ is kept in an electric field $\vec{E} = 2\,\hat{i} + 4\,\hat{j} + 7\,\hat{k}$. How much electric flux will come out through this surface?
a) 40 unit b) 50 unit c) 30 unit d) 20 unit

3. When the current changes from $+2A$ to $-2A$ in $0.05\ s$, an emf of $8\ V$ is induced in a coil. The co-efficient of self-induction of the coil is
a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H

4. The resistor of resistance R is connected to $25\ V$ supply and heat produced in it is $25\ J\,s^{-1}$. The value of R is
a) 225 Ω b) 1 Ω c) 25 Ω d) 50 Ω

5. Which of the following electromagnetic radiation is used for viewing objects through fog?
a) microwave b) gamma rays c) X- rays d) infrared rays

6. The current is flowing in south direction along a power line. The direction of magnetic field above the power line is
a) South b) East c) West d) North

7. If the velocity and wavelength of light in air is V_a and λ_a and that in water is V_w and λ_w, then the refractive index of water is,
a) $\frac{V_w}{V_a}$ b) $\frac{V_a}{V_w}$ c) $\frac{\lambda_w}{\lambda_a}$ d) $\frac{V_a}{V_w}\frac{\lambda_a}{\lambda_w}$

8. A coil having an area A_o is placed in a magnetic field which changes from B_o to $4B_o$ in time interval t. The emf induced in the coil will be
a) $\frac{3A_oB_o}{t}$ b) $\frac{4A_oB_o}{t}$ c) $\frac{3B_o}{A_ot}$ d) $\frac{4B_o}{A_ot}$

9. In an electron microscope, the electrons are accelerated by a voltage of $14\ kV$. If the voltage is changed to $224\ kV$, then the de Broglie wavelength associated with the electrons would
a) increase by 2 times b) decrease by 2 times
c) decrease by 4 times d) increase by 4 times

10. In Young's Double slit experiment fringe width decreases with increase in
a) λ b) D c) d d) None of these

11. The mass of a ^7_3Li nucleus is $0.042\ u$ less than the sum of the masses of all its nucleons. The average binding energy per nucleon of ^7_3Li nucleus is nearly
a) 46 MeV b) 5.6 MeV c) 3.9 MeV d) 23 MeV

12. A photon in motion has a mass
a) $\frac{c}{h\nu}$ b) $\frac{\nu}{h}$ c) $h\nu$ d) $\frac{h\nu}{c^2}$

13. The Zener diode is primarily used as
a) Rectifier b) Amplifier c) Oscillator d) Voltage regulator

14. For a transistor, in a common emitter arrangement, the alternating current gain (β) is given by
a) $\beta = \left[\frac{\Delta I_C}{\Delta I_B}\right]_{V_c}$ b) $\beta = \left[\frac{\Delta I_B}{\Delta I_C}\right]_{V_c}$ c) $\beta = \left[\frac{\Delta I_C}{\Delta I_E}\right]_{V_c}$ d) $\beta = \left[\frac{\Delta I_E}{\Delta I_C}\right]_{V_c}$

15. The alloys used for muscle wires in Robots are
a) Shape memory alloys b) Gold copper alloys
c) Gold silver alloys d) Two dimensional alloys

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$6 \times 2 = 12$

PART – II

Answer any six questions. Question no. 23 is compulsory:

16. What is 'corona discharge'?
17. Current is a scalar quantity. Why?
18. The repulsive force between two magnetic poles in air is 9×10^{-3} N. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate the pole strength of each pole.
19. State Fleming's right hand rule.
20. Why does sky appears blue colour?
21. What is diffraction?
22. How will you define threshold frequency?
23. Find the angular momentum of the electron revolving in the 5th orbit of hydrogen atom? ($h = 6.6 \times 10^{-34}$ Js)
24. Define forbidden energy gap?

$6 \times 3 = 18$

PART – III

Answer any six questions. Question no. 31 is compulsory:

25. What are the differences between soft and hard ferromagnetic materials?
26. Define coefficient of self-induction on the basis of magnetic flux and induced emf
27. What are Fraunhofer lines? How are they useful in the identification of elements present in the Sun?
28. Obtain the relation between focal length (f) and radius of curvature (R) of the spherical mirror.
29. A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm. (a) Calculate the capacitance of this capacitor. (b) If a 10 V battery is connected to the capacitor, what is the charge stored in any one of the plates? ($\varepsilon_o = 8.85 \times 10^{-12}\ N^{-1}m^{-2}C^2$)
30. What is polarizer and analyzer?
31. The work function of potassium is 2.30 eV. UV light of wavelength 3000 Å and intensity 2 W m^{-2} is incident on the potassium surface. Determine the maximum kinetic energy of the photo electrons
32. Give the properties of cathode rays.
33. Prove the following Boolean expressions using the laws and theorems of Boolean algebra.
    $(i)(A+B)(A+\bar{B}) = A$ $(ii)\ A(\bar{A}+B) = AB$ $(iii)(A+B)(A+C) = A+BC$

$5 \times 5 = 25$

PART – IV

Answer all the questions:

34. a) How the emf of two cells is compared using potentiometer?

    (Or)

    b) Prove laws of reflection using Huygens principle.

35. a) Describe the principle, construction and working of Cyclotron.

    (Or)

    b) Calculate the electric field due to a dipole on its equatorial line.

36. a) Obtain the law of radioactive decay.

    (Or)

    b) Write down Maxwell equations in integral form.

37. a) Give the construction and working of photo emissive cell.

    (Or)

    b) Explain the construction and working of a full wave rectifier.

38. a) (i) Define capacitive reactance.

    (ii) A capacitor of capacitance $\frac{10^2}{\pi}\ \mu F$ is connected across a 220 V, 50 Hz A.C. mains. Calculate the capacitive reactance and RMS value of current.

    (Or)

    b) (i) Define dispersive power of the prism.

    (ii) Find the dispersive power of a prism if the refractive indices of flint glass for red, green and violet colours are 1.613, 1.620 and 1.632 respectively.

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